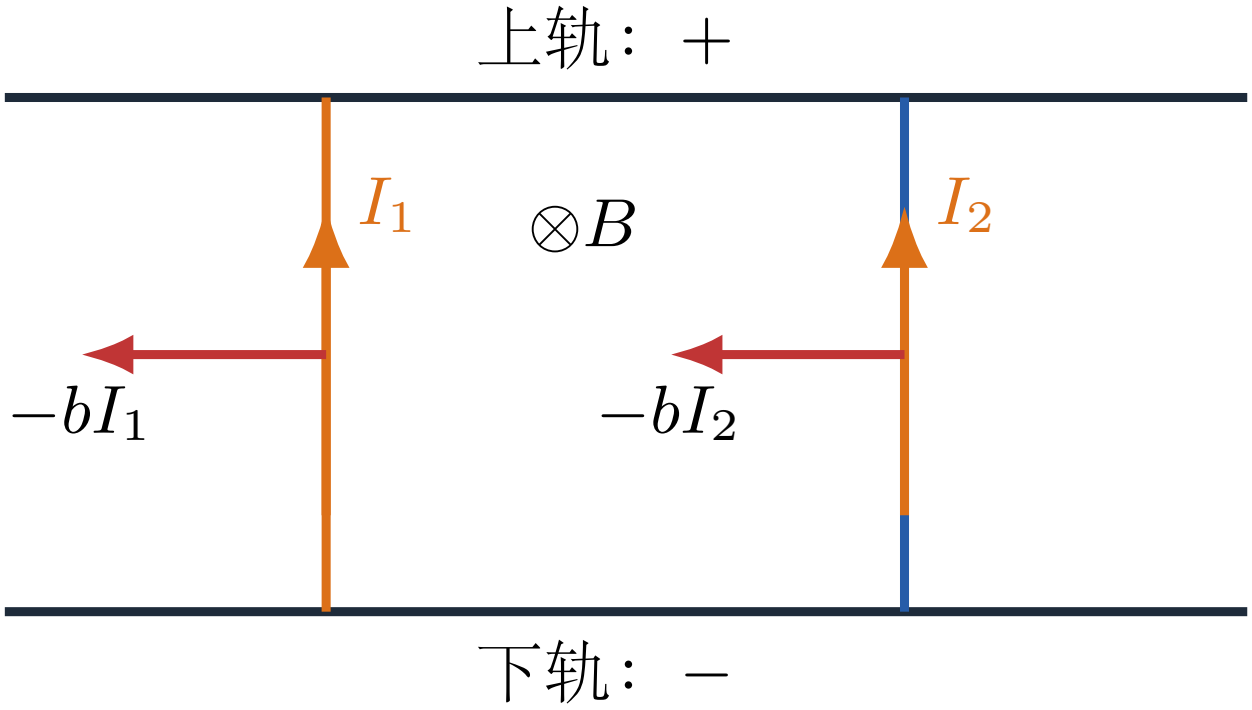

$$r_i I_i = bv_i - V$$

$$m_i \dot{v}_i = -bI_i$$

$$C\dot{V} = I_1 + I_2$$

$$m_1 v_1 + m_2 v_2 + bCV = \text{const}$$